Adapting IDS to newly labelled attacks

IDS · Classifying network flows as benign or attack types

Initial deployment

Labelled benign traffic
and known attacks

Newly labelled attack

A small labelled sample
of a newly observed attack

Detector update

Recognize the new attack
and preserve earlier quality

XGBoost Comparator: refit on all labelled examples available so far

TabPFN Fixed pretrained weights; labelled examples supplied as context

Comparison · New-attack detection · Retained quality · Update cost

TabPFN · Weights, context and query

Prediction rules pretrained on synthetic tabular tasks · TabPFN-3, arXiv 2026

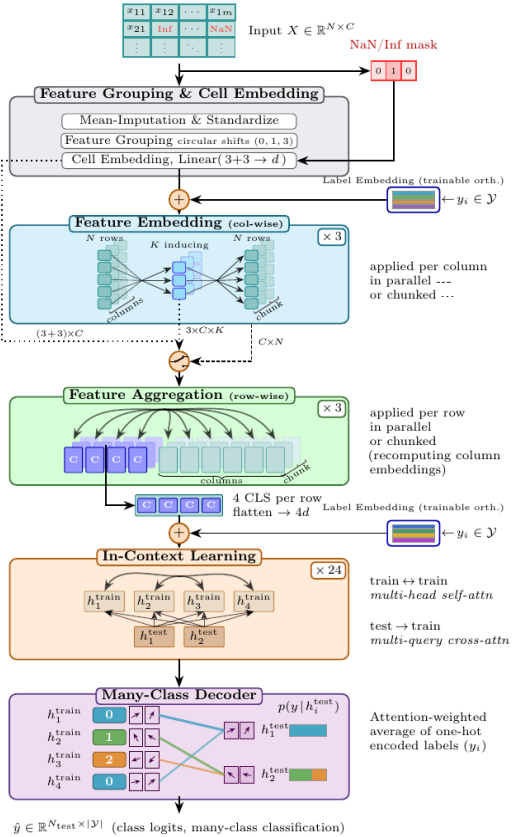


In-context learning Predict the query using its context, with no weight optimization

Fine-tuning Split IDS examples into context/query; update weights by backpropagating the labelled-query loss

Current comparison · Original pretrained weights, with no IDS fine-tuning

v3 architecture · Representing network flows



1 Encode values · Cell embedding

Convert each value, such as packet length, into a vector
Include an indicator for missing values

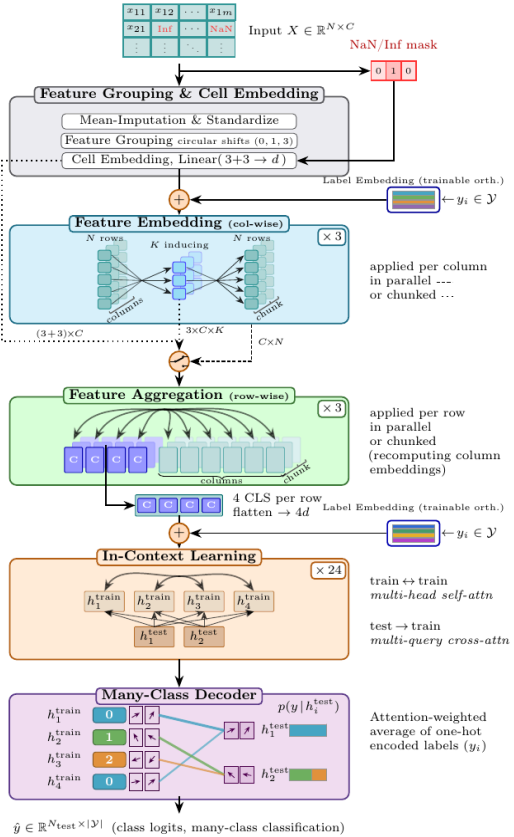
2 Summarize columns · Feature embedding

Summarize each feature across multiple flows
Use the labels supplied with the context

3 Combine features · Feature aggregation

Combine a flow's features into one representation
Prepare it for comparison with other flows

v3 architecture · Context-based prediction



4 Consult examples · In-context learning

Compare the query with labelled context examples
Weight their relevance through attention

5 Predict probabilities · Many-class decoder

Combine context labels with attention weights
Keep examples of both existing and new classes

Context updates also require computation

Preprocessing · Representations · Cache preparation
Measure update time separately from weight training